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Fusion Design Tradeoffs in Multimodal Models: Contradiction, Hallucination, and Robustness

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and Robustness**

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To Dana from 2022, you survived it

Abstract

Multimodal systems can exhibit reliability failures such as hallucination, contradiction sensitivity, and modality dominance, yet these behaviours are often studied separately. This thesis develops a conceptual framework for analysing how multimodal fusion architectures shape downstream reliability. The framework organises fusion design across five dimensions: fusion stage, interaction mechanism, representation and alignment, adaptivity, and computational cost. The framework is examined through a controlled contradiction experiment comparing LLaVA, which uses late fusion, with BLIP-2, which uses intermediate fusion through a Q-Former bottleneck. Both models are evaluated on 1,000 examples from the SNLI-VE dataset under matched entailment and contradiction conditions. LLaVA shows relatively balanced performance, achieving accuracies of 0.938 for entailment and 0.910 for contradiction. BLIP-2 displays a pronounced asymmetry, achieving 0.260 for entailment and 0.832 for contradiction. This result indicates that the two fusion architectures are associated with qualitatively different reliability behaviours, particularly an entailment collapse in BLIP-2. The findings support the use of fusion architecture as an important lens for understanding multimodal reliability. They also motivate design directions including contradiction-aware fusion, adaptive bottlenecks, and modality confidence estimation. The thesis contributes a unified analytical framework, empirical evidence linking fusion design to contradiction handling, and targeted recommendations for more reliable multimodal systems.

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Chapter 1

Introduction

Multimodal models integrate information from different modalities, such as vision and language, to perform tasks that require cross-modal understanding. However, the design of multimodal fusion remains largely guided by heuristics, requiring developers to balance performance, robustness, and complexity (Liang et al., 2023). When combining a language model with a vision encoder, decisions about how and where information should be integrated are often based on architectural precedent rather than a unified design principle (An et al., 2025).

These design choices are relevant to the reliability of multimodal systems. Existing research has identified problems including hallucination (Huang et al., 2024), robustness to missing or conflicting information (Ma et al., 2022), modality imbalance (Fan et al., 2023), and fairness (Alvarez et al., 2024). However, these phenomena are often studied separately, despite their potential relationship to how information from different modalities is represented and combined. This motivates examining multimodal reliability from the perspective of fusion design itself.

Accordingly, this thesis addresses the following research question: *How do fusion design choices affect a model’s ability to handle cross-modal contradiction, and what does that reveal about the tradeoffs inherent in multimodal fusion design?*

To investigate this question, the thesis proposes a conceptual framework for reasoning about multimodal fusion design and uses a controlled contradiction experiment to examine one aspect of that framework empirically. The experiment compares two multimodal architectures representing different fusion strategies and evaluates their behaviour when visual and textual information either agree or contradict one another. The findings are then used to identify broader implications for multimodal reliability and future fusion design.

The contribution of this thesis is therefore twofold: it provides a structured perspective for analysing fusion design choices in relation to multimodal reliability, and it provides an empirical case study illustrating how different architectural approaches can exhibit different behaviours under cross-modal contradiction. The framework is intended as an analytical heuristic rather than a predictive model, and its full formulation is presented in Chapter 3.

The empirical scope is deliberately limited. The study evaluates two models on a single visual entailment task and does not attempt to establish a general

ranking of multimodal architectures or isolate fusion design as the sole cause of the observed behaviours. Rather, the experiment serves as a focused investigation of the relationship between fusion architecture and contradiction handling.

The remainder of the thesis is organised as follows. **Chapter 2** reviews the relevant literature on multimodal fusion and reliability. **Chapter 3** introduces the proposed conceptual framework and derives the hypotheses examined in the empirical study. **Chapter 4** describes the experimental methodology, while **Chapter 5** presents the results. **Chapter 6** interprets these findings in relation to the framework and discusses their broader implications. Finally, **Chapter 7** summarises the contributions and limitations of the thesis and outlines directions for future work.

Chapter 2

Literature Review

2.1 Introduction to Multimodal Learning and Fusion

2.1.1 Multimodal Learning Foundations

Multimodal learning is a branch of Artificial Intelligence that is concerned with the joint processing and integration of information from multiple data modalities. Unlike unimodal systems, which operate on a single input type, multimodal systems combine complementary information sources to achieve richer, more context-aware representations. The field has grown substantially with the development of Deep Learning and Large Multimodal Models (LMMs) capable of jointly reasoning across heterogeneous forms of input modalities (Liang et al., 2023).

Common modalities include text, images, audio, video, sensor streams, and structured numerical data. Each possesses distinct structural and statistical properties: text is symbolic and sequential, images are spatial and continuous, and audio and sensor data are primarily temporal. This heterogeneity introduces fundamental challenges for representation learning, cross-modal alignment, synchronisation, and fusion.

Multimodal learning is applied across a wide range of tasks and domains. Common tasks include image captioning, visual question answering, speech recognition, multimedia retrieval, emotion recognition, autonomous driving, and human-computer interaction. More recently, multimodal approaches have become central to LMMs, where the objective extends beyond simple prediction toward cross-modal reasoning, contextual understanding, and interactive generation.

2.1.2 Evolution of Multimodal Systems

The development of multimodal systems has progressed through several major stages, beginning with classical machine learning approaches relying on handcrafted feature extraction and simple concatenation for modality combination. With the rise of Deep Learning, multimodal architectures adopted specialised neural encoders — convolutional neural networks (CNNs) for images and recurrent neural networks (RNNs) or long short-term memory networks (LSTMs) for sequential data

— significantly improving representation learning and cross-modal interaction.

More recently, Transformer-based architectures and attention mechanisms (Vaswani et al., 2017) enabled the development of LMMs, where multiple modalities are processed within unified token-based frameworks and shared embedding spaces. Representative multimodal systems include CLIP (Radford et al., 2021), Flamingo (Alayrac et al., 2022), BLIP-2 (Li et al., 2023), LLaVA (Liu et al., 2023), and modern commercial multimodal assistants such as Gemini (Gemini, 2023) and GPT-4o (OpenAI, 2024).

2.1.3 The Multimodal Fusion Problem

These structural differences, introduced in Section 2.1.1, become especially consequential under fusion, where representations from fundamentally different information spaces must interact meaningfully. As a result, multimodal systems must operate across fundamentally different information spaces while maintaining meaningful cross-modal relationships.

A further difficulty arises from semantic inconsistency and cross-modal misalignment: information from different modalities may partially overlap, contradict one another, or provide incomplete contextual signals. Additionally, multimodal systems frequently encounter noisy inputs, corrupted data, missing modalities, and varying levels of modality reliability (Ma et al., 2022). In practical environments, certain modalities tend to dominate the decision-making process due to stronger signal quality, dataset bias, or architectural design choices (Fan et al., 2023), which potentially lead to unstable reasoning and reduced robustness.

Consequently, multimodal fusion should not be understood solely as a feature combination problem. Rather, it simultaneously encompasses representation learning, cross-modal interaction modelling, reliability estimation, and computational optimisation. Effective fusion strategies must therefore balance semantic alignment, contextual reasoning, uncertainty handling, and computational efficiency while preserving robustness under imperfect multimodal conditions.

2.1.4 Conceptual Framework of Fusion Design

The complexity outlined above suggests that multimodal fusion should not be interpreted as a single architectural decision, but rather as a multidimensional design problem shaped by multiple interacting factors. To structure the analysis presented throughout this thesis, a conceptual analytical framework is introduced to describe the major dimensions influencing multimodal fusion behaviour and reliability:

$$F = f(S, I, R, A, C, \text{Task}, \text{Domain})$$

where S represents the Fusion Stage, I denotes the Interaction Mechanism between modalities, R corresponds to the Representation and Alignment space, A refers to Adaptivity and Reliability Awareness, and C captures Computational cost and Efficiency Constraints. The additional dimensions of task and application

domain are included to acknowledge that multimodal fusion strategies may behave differently depending on contextual requirements and operational environments.

This formulation defines a conceptual model of multimodal fusion rather than a formal mathematical model or optimisation objective. It represents fusion architectures through five interacting design dimensions and provides a structured basis for describing, comparing, and analysing how these dimensions may relate to reliability behaviour. The model does not specify a quantitative relationship between the dimensions or identify an optimal architecture. The framework reflects the observation that fusion-related behaviours — robustness, contradiction handling, hallucination susceptibility, modality dominance, and computational scalability — rarely emerge from a single architectural component alone. Rather, they arise from the interaction between representation strategies, cross-modal reasoning mechanisms, reliability estimation approaches, and system-level constraints.

The proposed framework constitutes the central analytical structure of this thesis and is used throughout the subsequent methodology, discussion, and conclusion chapters. In particular, the framework provides a systematic lens for examining how different multimodal fusion strategies influence reliability, robustness, interpretability, and reasoning consistency under challenging multimodal conditions.

2.2 Fusion Architectures and Interaction Mechanisms

2.2.1 Classical Fusion Taxonomy

Multimodal fusion strategies are traditionally organised into three approaches — early, intermediate, and late fusion — defined by the stage at which modalities are combined. Although this taxonomy remains a useful starting point, the complexity of Transformer-based architectures and LMMs has exposed its limitations.

Early fusion integrates raw inputs or low-level features into a shared representation space from the outset, enabling direct cross-modal correlation at the cost of sensitivity to noise, synchronisation mismatches, and missing modalities. Late fusion takes the opposite approach, processing modalities independently and combining outputs only at the final prediction stage. This improves modularity and fault tolerance but limits deeper semantic interaction between modalities. Intermediate fusion occupies the space between these extremes, allowing modalities to interact progressively across hidden representation layers and supporting richer cross-modal reasoning and hierarchical semantic integration. Hybrid architectures extend this further by combining multiple fusion strategies within the same system to balance robustness, flexibility, and representational depth (Poria et al., 2016).

Despite their continued utility, these categories are insufficient for describing Transformer-era multimodal systems. Modern LMMs employ attention-based interaction mechanisms, shared token spaces, dynamic routing, and adaptive modality weighting in ways that blur classical boundaries (An et al., 2025). Contemporary architectures are therefore better understood through the nature of cross-modal interaction — representation alignment, reliability adaptation, and contextual reasoning — rather than the stage at which fusion nominally occurs.

2.2.2 Attention and Cross-Attention Mechanisms

The Transformer architecture (Vaswani et al., 2017) introduced attention mechanisms as a central component of modern multimodal learning systems. Attention enables models to dynamically weight input components by relevance, allowing multimodal systems to selectively prioritise information across modalities rather than relying on fixed fusion rules. In Transformer-based architectures, self-attention mechanisms model relationships within the same modality by learning dependencies between tokens, image patches, audio segments, or other internal representations, capturing long-range contextual relationships and hierarchical semantic structures within individual modalities.

Cross-attention mechanisms extend this principle by enabling directed interactions between modalities. In cross-attention, representations from one modality influence the processing of another by acting as contextual queries, keys, or values during attention computation. A useful concrete example: when a model processes the question “what colour is the car on the left?”, cross-attention allows the textual query to direct visual attention toward the relevant image region rather than processing the entire scene uniformly. Visual information may similarly condition language generation in image captioning or visual question answering, establishing semantic grounding between modalities. In practice, this mechanism is often augmented with gated residual connections, which regulate how much cross-modal information is incorporated at each layer — allowing the model to selectively absorb or suppress visual signals depending on contextual relevance rather than passing all attended information forward unconditionally.

These architectural principles have been operationalised in distinct ways across modern LMMs. The Querying Transformer (Q-Former) introduced in BLIP-2 (Li et al., 2023) mediates vision-language interaction through a set of learned query vectors that attend to frozen visual encoder representations, effectively acting as a trainable bottleneck that extracts task-relevant visual information before passing it to the language model. This design deliberately constrains cross-modal information flow, trading representational completeness for computational efficiency and training stability. The Perceiver Resampler used in Flamingo (Alayrac et al., 2022) adopts a related strategy, using a fixed number of latent queries to compress variable-length visual feature sequences into compact representations that can be consumed by a frozen language model through cross-attention layers. Both architectures reflect a broader design principle: rather than allowing unrestricted interaction between visual and linguistic representations, intermediate bottleneck structures selectively compress and route cross-modal information, shaping which visual signals ultimately influence language generation (Nagrani et al., 2021).

Attention-based architectures therefore enable contextual weighting, where the relative importance of modalities shifts dynamically depending on task requirements, input quality, or environmental conditions. However, because attention mechanisms directly determine how modality influence is distributed throughout a model, their design is also a primary driver of modality dominance, cross-modal imbalance, and hallucination risk — particularly when one modality consistently attracts

disproportionate attention weight under noisy or contradictory conditions. This connection between attention design and reliability failure is examined further in Section 2.3.

2.2.3 Modern Hybrid and Adaptive Fusion

Contemporary multimodal architectures increasingly rely on hybrid and adaptive fusion strategies that combine multiple interaction mechanisms within the same system rather than adhering strictly to classical fusion approaches. Modern LMMs frequently integrate techniques such as gated fusion, bottleneck fusion, dynamic modality weighting, shared embedding or token spaces, and attention-based cross-modal routing to balance representational richness, computational efficiency, and robustness. Gated fusion mechanisms selectively regulate modality influence, bottleneck architectures constrain information exchange through compressed interaction layers, and dynamic weighting approaches adapt modality importance according to contextual relevance or estimated reliability. At the same time, shared token-space architectures enable modalities to interact within unified Transformer frameworks where text, image patches, audio segments, and other representations are processed through common attention mechanisms (Wang et al., 2024). Consequently, modern multimodal systems are best understood not as single fusion architectures, but as compromise-driven hybrid ecosystems designed to balance competing objectives including semantic alignment, scalability, robustness, reasoning flexibility, and computational cost.

2.3 Cross-Modal Contradiction and Hallucination

2.3.1 Cross-Modal Contradiction

Cross-modal contradiction refers to cases in which different modalities provide semantically inconsistent, conflicting, or mutually incompatible information within the same multimodal context. Such contradictions may arise explicitly, where modalities directly oppose one another, or implicitly, where contextual interpretations become semantically incompatible during multimodal reasoning. For example, an image may depict a crowded urban environment while the accompanying text describes an isolated rural setting, or an audio signal may contradict the emotional content expressed through facial expressions in a video sequence. These inconsistencies introduce substantial challenges for multimodal systems because fusion mechanisms are generally optimised to integrate complementary information rather than resolve disagreement between modalities.

The manner in which contradictions propagate through a multimodal system depends heavily on the selected Fusion Stage (S) and Interaction Mechanism (I) within the proposed conceptual framework. In early fusion architectures, contradictory information may become entangled at low-level representation stages, potentially causing noisy or unstable latent representations that affect the entire reasoning process downstream. Late fusion systems, in contrast, isolate modality-specific

processing until the final decision stage, which may improve robustness to localised contradictions but often limits deeper cross-modal reasoning and semantic reconciliation. Intermediate and hybrid fusion architectures introduce more complex propagation dynamics because modalities interact iteratively throughout hidden representation layers, allowing contradictions to be either progressively resolved through contextual refinement or amplified through recursive cross-modal influence.

Attention-based interaction mechanisms further complicate contradiction handling by dynamically redistributing modality importance during inference. In Transformer-based systems, dominant modalities may override weaker ones through attention allocation patterns, particularly when training distributions encourage reliance on learned statistical priors. Consequently, cross-modal contradiction is not merely a data inconsistency problem but also a fusion-design problem shaped by representation alignment, interaction structure, modality weighting behaviour, and reliability adaptation mechanisms. Understanding how contradictions propagate across different fusion architectures therefore becomes essential for evaluating multimodal robustness, reasoning consistency, and hallucination susceptibility in LMMs.

2.3.2 Hallucination in Multimodal Systems

Hallucination in multimodal systems refers to the generation of outputs, predictions, or reasoning processes that are not sufficiently supported by the available multimodal evidence (Bai et al., 2024). In the context of LMMs, hallucinations frequently emerge as fabricated grounding, where the system produces semantically plausible but factually unsupported descriptions, objects, relationships, or contextual interpretations. Unlike conventional prediction errors, multimodal hallucinations are particularly problematic because generated outputs often appear coherent and contextually convincing despite lacking reliable correspondence with the underlying modalities. As a result, hallucination represents not only a performance limitation but also a significant reliability and trustworthiness concern in multimodal reasoning systems (Huang et al., 2024).

Multimodal hallucinations may originate from several interacting architectural and representational factors. One major source is representation mismatch, where modalities are imperfectly aligned within shared embedding spaces, causing the system to infer semantic relationships that are weak, ambiguous, or nonexistent. Cross-modal inconsistencies, noisy inputs, incomplete contextual information, and insufficient grounding between modalities can further amplify this problem. Additionally, modern Transformer-based architectures heavily depend on attention mechanisms for contextual reasoning and modality interaction. While attention enables adaptive multimodal integration, it may also lead to attention overtrust, where the system disproportionately relies on dominant modalities, learned statistical priors, or high-frequency training patterns during inference. In such situations, the model may prioritise internally learned priors over actual multimodal evidence, generating outputs that reflect training distribution biases rather than the presented inputs.

The propagation of hallucinations is also strongly influenced by fusion architecture design. Early fusion systems may distribute noisy or misleading signals throughout shared representations at initial processing stages, while intermediate fusion architectures may recursively reinforce unsupported cross-modal associations through repeated interaction layers. In LMMs, hallucinations often emerge from the interaction between representation alignment errors, adaptive attention allocation, modality imbalance, and probabilistic language generation dynamics. Consequently, hallucination should be understood as a critical property of multimodal interaction mechanisms rather than as an isolated failure of a single modality or component. This perspective further reinforces the importance of reliability-aware fusion strategies capable of explicitly modelling uncertainty, contradiction, and modality confidence during multimodal reasoning.

2.3.3 Modality Overtrust and Semantic Dominance

A recurring challenge in multimodal systems is the tendency of certain modalities to dominate the fusion and reasoning process disproportionately. Modality overtrust refers to situations in which a multimodal model consistently prioritises one modality while underutilising or overriding signals from others. This behaviour emerges because modalities differ in representational capacity, signal quality, statistical prevalence, or predictive reliability within the training distribution. Consequently, multimodal systems may learn implicit modality hierarchies in which certain modalities become systematically privileged during inference and decision-making processes (Fan et al., 2023).

In LMMs, language frequently becomes the dominant modality due to the scale and statistical strength of pretrained language models integrated within multimodal architectures. Because language models already encode extensive semantic priors and world knowledge, visual, auditory, or sensory information may serve primarily as additional conditioning signals rather than equal contributors to reasoning. As a result, multimodal systems may generate outputs that remain linguistically plausible even when non-textual modalities provide contradictory or insufficient evidence. This phenomenon is particularly visible in image captioning, visual question answering, and multimodal dialogue systems, where generated responses may reflect dominant linguistic expectations rather than faithful multimodal grounding.

Over time, excessive dependence on dominant modalities can produce what this thesis terms *semantic collapse* — a process whereby multimodal representations gradually converge toward the interpretational biases of the strongest modality. Under these conditions, weaker modalities contribute progressively less meaningful information to the final representation space, reducing the diversity and complementarity that multimodal fusion is intended to achieve. Semantic collapse not only weakens robustness and contradiction sensitivity but may also amplify representational bias and uneven modality treatment across different tasks and environments. Consequently, modality dominance should be understood as both a reliability problem and a fairness-related issue, since unequal modality influence may systematically disadvantage certain forms of information during multimodal reasoning. This con-

nection becomes particularly important when considering fairness, bias propagation, and representational imbalance in multimodal systems, which are discussed further in Section 2.5.

2.3.4 Contradiction as a Lens for Reliability Evaluation

Cross-modal contradiction scenarios provide a particularly valuable perspective for evaluating the reliability and robustness of multimodal systems because they directly expose how fusion architectures behave under semantic tension and uncertainty. While many multimodal benchmarks focus primarily on predictive accuracy under coherent and well-aligned inputs, contradictory multimodal conditions force systems to reveal underlying assumptions regarding modality trust, attention allocation, representation alignment, and conflict resolution. In such settings, multimodal systems can no longer rely solely on complementary information integration and must instead determine how conflicting evidence should be interpreted, prioritised, or reconciled during reasoning.

Different fusion architectures respond to contradiction in substantially different ways. Early fusion systems may diffuse contradictory signals throughout shared representations, intermediate fusion architectures may recursively amplify inconsistent semantic interactions, while late fusion approaches may isolate disagreement at the decision level without performing deeper reconciliation. Similarly, attention-based interaction mechanisms may either improve contradiction sensitivity through adaptive contextual weighting or intensify modality dominance by disproportionately privileging historically stronger modalities. Consequently, contradiction scenarios act as stress tests that reveal hidden weaknesses in fusion-stage design, interaction mechanisms, modality weighting behaviour, and reliability adaptation strategies.

From this perspective, contradiction should not be viewed solely as an edge case or adversarial anomaly but rather as a natural evaluation framework for studying multimodal reliability. Contradictory conditions allow researchers to analyse whether multimodal systems preserve semantic grounding, detect inconsistency, maintain robustness under uncertainty, and avoid hallucination when modalities disagree. This thesis therefore treats cross-modal contradiction as a central analytical lens for examining multimodal fusion behaviour, directly motivating the empirical evaluation and comparative analysis presented in later chapters.

2.4 Robustness Under Noisy or Missing Modalities

Robustness under noisy or missing modalities represents one of the central challenges in multimodal learning, particularly in real-world environments where multimodal inputs are rarely perfectly synchronised, complete, or reliable. Noise may arise from sources including corrupted images, degraded audio quality, sensor malfunction, transcription errors, occlusion, or environmental interference, while missing modalities occur when one or more information sources become partially or entirely unavailable during inference (Ma et al., 2022). Because multimodal systems depend on interactions between heterogeneous data sources, the degradation of a

single modality can significantly destabilise the quality and consistency of multimodal reasoning.

The impact of noisy or missing modalities depends strongly on fusion architecture. Early fusion systems are generally most vulnerable: integrating modalities at low-level representation stages means corrupted or incomplete information propagates throughout the shared latent space, potentially destabilising the entire downstream reasoning process. Late fusion systems demonstrate greater resilience since each modality is processed independently prior to decision aggregation, allowing individual modalities to fail without corrupting the full representation pipeline — though at the cost of reduced capacity for deeper semantic interaction and contextual reconciliation between modalities.

Intermediate and hybrid fusion architectures introduce more complex robustness behaviours. Attention-based systems may dynamically reweight modalities according to contextual relevance, estimated confidence, or perceived signal quality, allowing models to partially compensate for degraded inputs during inference. However, this adaptivity is double-edged: if a system incorrectly estimates modality reliability or disproportionately overtrusts a dominant modality under uncertainty, adaptive mechanisms may amplify instability rather than contain it. Robustness in multimodal systems is therefore not determined solely by modality availability, but by how fusion architectures distribute, regulate, and adapt modality influence throughout the reasoning process. These challenges connect directly to the Adaptivity dimension (*A*) of the proposed conceptual framework.

It is worth illustrating concretely what reliability-aware fusion looks like in practice. One well-established approach is confidence-weighted fusion, in which each modality is assigned a dynamic reliability score — derived from prediction entropy, dropout-based uncertainty sampling, or learned confidence estimation — and modality contributions to the final representation are scaled accordingly. Under this scheme, a severely degraded visual input receives lower weight during fusion while the textual modality compensates proportionally, rather than the system treating all modalities as equally trustworthy regardless of input quality. A related strategy, uncertainty-aware fusion (Han et al., 2022), explicitly models the epistemic uncertainty of each modality using evidential Deep Learning, allowing the system to quantify how much confidence to place in each information source before integration. Both approaches operationalise the (*A*) dimension by treating modality reliability as a dynamic variable rather than a fixed architectural assumption.

The practical importance of such mechanisms grows as multimodal systems expand into safety-critical domains — medical diagnosis, autonomous driving, human-robot interaction — where the consequences of miscalibrated modality trust are severe. The discussion chapter returns to adaptivity as a design dimension in greater depth, examining how reliability-aware fusion strategies interact with contradiction handling and modality dominance under challenging multimodal conditions.

2.5 Fairness, Bias, and Modality Dominance

Fairness and bias in multimodal systems are increasingly recognised as fundamental challenges rather than secondary ethical concerns (Alvarez et al., 2024). Because multimodal architectures integrate information originating from heterogeneous modalities with unequal representational properties, fusion mechanisms inevitably shape how information is emphasised, interpreted, and propagated during reasoning. As a result, multimodal systems may inherit, expand, or redistribute biases not only from training data but also from the architectural structure of the fusion process itself. Consequently, multimodal fairness cannot be understood solely through dataset composition or output evaluation; it must also be analysed as a property of fusion design and modality interaction.

One major source of multimodal bias emerges through modality dominance, where certain modalities systematically exert greater influence over the final representation and decision-making process. In many LMMs, language becomes the dominant modality due to the scale and semantic strength of pretrained large language models that are part of the multimodal models, while visual, auditory, or sensory modalities act primarily as supporting contextual signals. This imbalance may cause multimodal systems to centre on dominant modalities even when weaker modalities provide contradictory, corrective, or contextually important information. Such behaviour can suppress modality diversity, reduce contradiction sensitivity, and produce semantically biased outputs shaped by historically overrepresented modalities (Fan et al., 2023).

Fusion architectures themselves may further increase representational bias through attention allocation, weighting strategies, alignment objectives, and interaction mechanisms. Attention-based systems disproportionately focus on statistically frequent modality patterns, and shared representation spaces often encode uneven semantic relationships between modalities shaped by training distribution characteristics. Adaptive fusion systems that estimate modality reliability can further reinforce existing modality hierarchies when reliability estimation is itself biased toward learned statistical priors. Consequently, multimodal bias often emerges as an interaction effect between training data, representation learning, and fusion architecture design rather than from isolated components alone.

From this perspective, fusion mechanisms should not be considered architecturally neutral. The manner in which modalities are aligned, weighted, fused, and dynamically prioritised directly shapes how multimodal systems interpret reality, resolve contradictions, and distribute semantic influence between information sources. Fairness therefore becomes a design consequence of multimodal interaction structures rather than an incidental issue appearing only at the output stage. Understanding how fusion architectures influence modality dominance and bias propagation is essential for developing multimodal systems that are not only accurate and efficient, but also robust, interpretable, and semantically balanced across diverse multimodal environments.

2.6 Research Gap and Thesis Positioning

2.6.1 Limitations of Current Literature

Current multimodal learning literature has made substantial progress in the study of fusion architectures, hallucination, robustness, fairness, and computational efficiency (Liang et al., 2023; Li and Tang, 2024). However, these problems are still predominantly investigated as relatively separate research directions with limited theoretical integration between them. Fusion-oriented studies largely focus on improving predictive accuracy, representation alignment, or cross-modal interaction efficiency (An et al., 2025), while hallucination research concentrates on output faithfulness and grounding failures in vision-language systems (Bai et al., 2024). Similarly, robustness research commonly examines adversarial perturbations, noisy inputs, or missing modality degradation (Ma et al., 2022; Xu et al., 2023), whereas fairness-related studies focus primarily on demographic bias and representational disparity within multimodal predictions (Alvarez et al., 2024).

Although these research areas frequently address related symptoms, the literature rarely treats them as interconnected consequences of shared architectural and representational design choices. Systematic reviews of multimodal learning consistently identify robustness limitations, hallucination behaviour, interpretability challenges, and fairness concerns as distinct problem categories rather than components of a unified analytical structure (Li and Tang, 2024).

Nevertheless, several emerging research directions provide partial evidence toward a more structurally connected understanding of multimodal reliability. Trusted multi-view and uncertainty-aware fusion approaches increasingly connect multimodal integration with robustness and reliability estimation by dynamically assessing modality quality under uncertain conditions (Han et al., 2022). Recent modality-conflict hallucination studies further suggest that hallucination may emerge not only from model-output mismatch but also from semantic conflict between modalities themselves. Similarly, research on spurious multimodal bias demonstrates that multimodal systems frequently rely on non-causal modality correlations, simultaneously producing brittle robustness behaviour and biased reasoning patterns. In parallel, fairness-oriented multimodal studies increasingly recognise that modality imbalance and representational asymmetry influence bias propagation within LMMs.

Despite these partial convergences, the literature lacks a unified analytical framework treating multimodal fusion architecture as a causal design variable shaping downstream system behaviour. Existing work approaches fusion either as a taxonomy of architectural categories or as a performance optimisation problem, and even studies labelled as fusion-centric primarily propose new mechanisms rather than analysing fusion design as an upstream structural factor governing hallucination, robustness, modality dominance, and fairness simultaneously. Fusion-stage selection, interaction mechanisms, representation alignment, and adaptive modality weighting continue to be treated as independent variables rather than interconnected determinants of reliability-related failure modes.

This thesis argues that multimodal fusion should be understood not merely as

a mechanism for combining modalities, but as an upstream architectural design space that governs how information influence is distributed throughout multimodal reasoning. Within this perspective, the Fusion Stage affects how early modality dominance can emerge; Interaction Mechanisms determine whether modalities reinforce or override one another; Representation Alignment influences semantic grounding and hallucination susceptibility; while adaptivity and uncertainty estimation shape robustness under noisy, missing, or contradictory conditions. Consequently, hallucination, robustness failure, fairness disparity, and modality dominance are interpreted not as isolated phenomena, but as interconnected downstream consequences of modality integration design choices.

2.6.2 Thesis Positioning

This thesis positions multimodal fusion as a multidimensional tradeoff space rather than a single architectural optimisation problem. Instead of treating hallucination, robustness, fairness, contradiction handling, and modality dominance as isolated system failures, the thesis approaches them as interconnected reliability behaviours emerging from multimodal interaction structure and fusion design choices. The proposed conceptual framework introduced in Section 2.1.4 is intended to provide an analytical lens for examining how Fusion Stage, Interaction Mechanisms, Representation Alignment, Adaptivity, and Computational Constraints collectively shape multimodal system behaviour.

The objective of this work is not to provide an exhaustive empirical benchmark across all multimodal architectures or datasets. Rather, the thesis aims to contribute a conceptual synthesis supported by targeted empirical analysis capable of illustrating how different fusion paradigms influence contradiction propagation, semantic grounding, robustness, and modality dominance under challenging multimodal conditions. In this sense, the contribution of the thesis is primarily analytical and structural: it seeks to connect research areas that are currently studied in parallel but rarely interpreted as manifestations of the same underlying fusion dynamics.

More specifically, the thesis argues that multimodal fusion mechanisms are not architecturally neutral. Fusion design determines how modality influence is distributed throughout the reasoning process, which modalities become dominant under uncertainty, how contradictions are resolved or suppressed, and how semantic grounding is maintained under incomplete or conflicting information. Consequently, reliability, robustness, and fairness are interpreted not as isolated evaluation metrics, but as systemic consequences of modality integration strategies.

2.6.3 Transition to Conceptual Framework

The literature reviewed throughout this chapter demonstrates that multimodal learning research has made substantial progress in fusion architectures, multimodal reasoning, robustness analysis, hallucination mitigation, and fairness-aware learning. However, these challenges remain predominantly investigated as relatively separate research directions despite increasing evidence that they emerge from in-

terconnected multimodal interaction dynamics and fusion design choices. Existing studies frequently optimise multimodal systems for performance improvements, while downstream behaviours such as contradiction sensitivity, hallucination, modality dominance, and robustness degradation are analysed independently within separate evaluation contexts.

This fragmentation motivates the need for a broader analytical perspective capable of examining multimodal fusion as a multidimensional design space rather than a single architectural decision. The next chapter therefore introduces a conceptual framework for reasoning about multimodal fusion tradeoffs across multiple interacting dimensions, including Fusion Stage, Interaction Mechanisms, Representation Alignment, Adaptivity, and Computational Constraints. One dimension of this framework — contradiction handling across different fusion paradigms — is subsequently operationalised empirically in Chapter 4 through targeted analysis of multimodal conflict behaviour and reliability outcomes.

Chapter 3

Conceptual Framework and Hypotheses

3.1 The Fusion Design Space

As introduced in Section 2.1.4, the proposed framework conceptualises multimodal fusion not as a single architectural choice but as a multidimensional design space shaped by multiple interacting factors:

$$F = f(S, I, R, A, C, \text{Task}, \text{Domain})$$

where S represents the fusion stage, I denotes the interaction mechanism between modalities, R corresponds to the representation and alignment space, A captures adaptivity and reliability awareness, and C represents computational cost and efficiency constraints. The dimensions of task and domain acknowledge that fusion effectiveness depends on application context and operational requirements. The framework functions as a conceptual analytical lens, not a predictive mathematical model; its central claim is that commonly studied multimodal reliability phenomena — contradiction sensitivity, hallucination susceptibility (Bai et al., 2024), robustness degradation (Ma et al., 2022), modality dominance (Fan et al., 2023), and fairness disparity (Alvarez et al., 2024) — are downstream consequences of design choices within this space.

To connect the framework to empirical testing, each dimension can be mapped to observable behavioural outcomes. The fusion stage S governs how cross-modal contradiction propagates through the system, determining whether inconsistencies are contained or compounded. The interaction mechanism I shapes whether modalities reinforce or compete with one another, directly influencing hallucination susceptibility under conflicting inputs. Representation and alignment R affect the likelihood of unsupported multimodal associations, as discussed in Section 2.3.2. Adaptivity A regulates robustness under noisy or contradictory conditions through confidence estimation and uncertainty modelling, a challenge examined in Section 2.4. Computational constraints C bound the feasible degree of cross-modal interaction. Of these dimensions, S and I are the most directly operationalisable through model selection

and controlled input manipulation, and therefore serve as the primary empirical targets of this thesis. Contradiction between modalities provides a controlled stress test — consistent with the argument developed in Section 2.3.4 — for exposing how these design dimensions influence system behaviour.

3.2 Tradeoff Predictions

The framework predicts that multimodal system behaviour emerges from a series of architectural tradeoffs rather than any single design decision. The following predictions, derived from the discussion of fusion architectures in Sections 2.2 and cross-modal contradiction in Section 2.3, provide the theoretical basis for the empirical analysis in Chapter 4.

3.2.1 Fusion Stage S

Prediction. Earlier fusion increases the risk that contradictory information propagates through shared representations, while later fusion improves contradiction containment at the cost of weaker cross-modal reasoning.

If fusion occurs early, shared representations are exposed to contradictory information before semantic reconciliation can occur, increasing the risk that inconsistencies propagate into downstream reasoning and contribute to hallucination. If fusion is deferred to the decision stage, contradiction propagation is limited and modality-specific robustness improves, but at the cost of reduced cross-modal reasoning depth and semantic verification. Intermediate and hybrid architectures are predicted to occupy a middle ground, with the balance between interaction richness and contradiction containment depending on the specific mechanisms employed. This architectural spectrum mirrors the classical fusion taxonomy reviewed in Section 2.2.1.

3.2.2 Interaction Mechanism I

Prediction. Stronger cross-modal interaction can improve semantic integration, but it may also amplify modality dominance when attention or weighting mechanisms over-prioritise historically stronger modalities.

If modalities interact through cross-attention, semantic integration improves through dynamic weighting, but modality dominance may be amplified when attention allocation concentrates on historically stronger signals, a risk that connects to the attention overtrust discussion in Section 2.3.3. If interaction is constrained through bottleneck or gated mechanisms, stability improves through regulated information exchange, potentially at the cost of representational richness. The interaction mechanism therefore determines whether cross-modal contradiction is resolved, suppressed, or amplified.

3.2.3 Representation and Alignment R

Prediction. Stronger cross-modal alignment improves grounding and reduces unsupported associations, while weak alignment increases the risk of contradiction

misinterpretation and fabricated grounding.

If cross-modal alignment is strong, the likelihood of unsupported associations decreases and semantic grounding improves. If alignment is weak or imperfect, representation mismatch increases, and systems are expected to infer relationships not supported by available evidence. This mechanism is closely tied to the hallucination sources identified in Section 2.3.2. Poorly aligned spaces therefore predict higher rates of contradiction misinterpretation and fabricated grounding.

3.2.4 Adaptivity and Reliability Awareness A

Prediction. Reliability-aware systems can improve robustness under noisy or contradictory conditions, but only when uncertainty estimation is well calibrated.

If a system incorporates confidence estimation or uncertainty modelling, robustness under noisy or contradictory conditions improves, but only when reliability assessment itself is well calibrated. If reliability estimation is inaccurate, adaptive mechanisms introduce a second-order risk: systems may overtrust dominant modalities and reinforce existing biases rather than compensating for them. This conditional benefit connects to the modality dominance challenges described in Section 2.3.3 and the robustness discussion in Section 2.4.

3.2.5 Computational Cost C

Prediction. Architectures that support richer cross-modal communication can improve integration quality, but they impose higher computational costs and may be less feasible in resource-constrained settings.

If an architecture supports extensive cross-modal communication and adaptive routing, integration quality improves at higher computational cost. If the architecture pursues efficiency through simpler fusion, nuanced reasoning and contradiction resolution are sacrificed. This tradeoff determines which fusion strategies are feasible in resource-constrained deployment environments.

3.2.6 Narrowing to Empirical Targets

The five dimensions above define a space of competing objectives rather than universally optimal solutions. Among them, fusion stage S and interaction mechanism I are selected as the primary empirical targets of this thesis. They are the most directly controllable through model selection: different architectures instantiate different positions in the S, I space, and cross-modal contradiction provides a controlled stress test that exposes their behavioural consequences.

The remaining dimensions, representation and alignment R , adaptivity and reliability awareness A , and computational cost C , are revisited theoretically in the discussion chapter, where their interactions with the empirical findings are examined in light of the literature reviewed in Chapter 2.

3.3 Hypotheses

Three hypotheses are derived from the framework predictions above. H1 and H2 are directly evaluated through the empirical comparison in Chapter 4. H3 is included as a framework-derived theoretical prediction; it is evaluated empirically only if a suitable reliability-aware model is available for comparison, and otherwise remains a motivated direction for future research. Each hypothesis is stated in falsifiable form, followed by the mechanistic rationale that motivates it.

H1 – Fusion Stage Hypothesis

Models employing intermediate fusion will exhibit greater performance degradation under cross-modal contradiction than models employing late fusion.

The rationale is architectural. In intermediate fusion systems, contradictory signals interact within shared representation layers throughout the reasoning process, allowing inconsistencies to propagate and compound before any decision is formed. Late fusion maintains modality-specific processing until the final stage, limiting the surface area over which contradiction can influence internal representations. This prediction directly extends the analysis of how fusion stage shapes contradiction propagation introduced in Section 2.3.1. H1 therefore predicts that the stage at which modalities are integrated is a primary determinant of contradiction sensitivity.

H2 – Interaction Mechanism Hypothesis

Models employing cross-attention-based interaction will exhibit lower hallucination rates under cross-modal contradiction than models relying on simpler fusion mechanisms.

Cross-attention enables selective, dynamic weighting of modality-specific information, allowing models to prioritise relevant signals and partially suppress contradictory ones during reasoning. Simpler or more static fusion mechanisms offer no equivalent selective capacity, making them more prone to propagating contradictory information through forced integration. However, this benefit is conditional: cross-attention reduces hallucination only when attention allocation does not systematically overtrust the dominant modality under conflict. This risk was discussed in Section 2.3.3 in relation to modality dominance and semantic collapse. H2 therefore predicts that interaction mechanism flexibility is a significant determinant of hallucination behaviour under contradiction, and the empirical test reveals whether selective weighting or modality dominance prevails in the architectures examined.

H3 – Adaptivity Hypothesis

Models incorporating explicit reliability-awareness mechanisms will maintain more stable performance across consistent and contradictory multimodal conditions than models lacking such mechanisms.

Reliability-aware fusion, through confidence-weighted integration or uncertainty-

aware modality weighting, allows systems to dynamically adjust modality influence in response to signal quality rather than treating all inputs as equally trustworthy. This capacity is predicted to reduce sensitivity to contradiction and modality-specific failure. H3 is the most theoretically motivated of the three hypotheses and connects most directly to the adaptivity dimension A of the framework. Its empirical evaluation is contingent on model availability.

3.4 Scope and Limitations

The three hypotheses are deliberately scoped to dimensions S and I as primary empirical targets, with A as a secondary prediction. The remaining dimensions — representation alignment R , computational cost C , task, and domain — are treated as theoretical variables discussed in Chapter 2 and revisited in the discussion chapter rather than as direct objects of empirical testing. This scope reflects a deliberate methodological decision appropriate to the scale of the thesis: rather than attempting shallow coverage of all dimensions, the empirical component offers targeted, controlled analysis of the dimensions most directly implicated in contradiction handling.

Together, the framework dimensions and hypotheses developed in this chapter establish the theoretical basis for the empirical investigation. H1 and H2 translate the fusion stage S and interaction mechanism I dimensions into testable predictions about how cross-modal contradiction affects system behaviour. The next chapter puts these predictions to the test through a controlled contradiction experiment that compares two models occupying distinct positions in the S, I design space: one predominantly late-fusion architecture and one intermediate-fusion architecture with cross-attention-based interaction. Chapter 4 details the task construction, experimental conditions, evaluation metrics, and analytical approach used to determine whether contradiction sensitivity and hallucination behaviour follow the trajectory predicted by the framework.

Chapter 4

Methodology

4.1 Framework Operationalisation

This chapter details the methodology used to empirically evaluate the conceptual framework proposed in Chapter 3. The experiment serves as an exploratory test of H1 and H2, the fusion stage and interaction mechanism hypotheses, focusing on how fusion design choices relate to model behaviour under cross-modal contradiction.

The experiment operationalises two of the five framework dimensions, Fusion Stage (S) and Interaction Mechanism (I), through the selection of design exemplars. The remaining dimensions, including Adaptivity (A), are addressed theoretically in Chapter 6. The mapping between framework variables and the experimental design is summarised in Table 4.1.

Table 4.1. Mapping of framework dimensions to experimental variables

Framework Dimension	Experimental Operationalisation
Fusion Stage (S)	Selection of models with late versus intermediate fusion
Interaction Mechanism (I)	Comparison of direct projection versus cross-attention bottlenecks
Contradiction	Construction of semantically conflicting image-hypothesis pairs in SNLI-VE

4.2 Model Selection as Design Exemplars

Rather than serving as a benchmark for ranking model performance, this experiment treats models as design exemplars that instantiate different positions in the (S, I) design space. Two open-source large multimodal models were selected for their architectural transparency and well-documented fusion mechanisms, allowing each model’s position in the design space to be established with confidence.

4.2.1 LLaVA-1.5-7B: Late Fusion Exemplar

LLaVA-1.5-7B (Liu et al., 2023) is selected as a representative of the late fusion paradigm. It uses a linear MLP projector to map visual features from a frozen CLIP-ViT-L/14 encoder (Radford et al., 2021) directly into the token space of the Vicuna-7B language model. Within the proposed framework, LLaVA occupies the late fusion and direct projection position. This architecture maintains modality specific processing until the transformer decoder’s deep layers, theoretically preserving high visual fidelity by avoiding early bottleneck compression.

4.2.2 BLIP-2-OPT-2.7B: Intermediate Fusion Exemplar

BLIP-2-OPT-2.7B (Li et al., 2023) is selected as a representative of the intermediate fusion paradigm. It mediates vision language interaction through a Querying Transformer, or Q-Former, which acts as a learned interaction mechanism between a frozen image encoder and a frozen language model, OPT-2.7B (Zhang et al., 2022). Within the framework, BLIP-2 occupies the intermediate fusion and cross-attention bottleneck position. The Q-Former compresses 196 visual patches into 32 query tokens via cross-attention, representing an architectural trade-off between representational richness and cross-modal interaction efficiency.

4.3 Dataset and Experimental Task

The experiment uses the SNLI-VE visual entailment dataset (Xie et al., 2019), a multimodal reasoning benchmark built on Flickr30k images (Plummer et al., 2015). The task is framed as a binary classification problem, Entailment versus Contradiction, to isolate the model’s sensitivity to semantic conflict.

- **Consistent Condition (Entailment):** Input pairs where the textual hypothesis is semantically supported by the visual premise.
- **Contradictory Condition (Contradiction):** Input pairs where the hypothesis directly conflicts with the visual premise.

A stratified subset of 1,000 examples, 500 entailment and 500 contradiction, was drawn from the SNLI-VE validation set to ensure balanced conditions for the exploratory analysis.

4.4 Experimental Environment and Constraints

The implementation was conducted in a Google Colab environment using a T4 GPU and 112GB of available disk space. These infrastructure constraints directly influenced several methodological decisions.

- **Quantization:** Both models were loaded in 4-bit quantization using `bitsandbytes`. While necessary to satisfy VRAM constraints, quantization is acknowledged

as a potential, though shared across both models, noise factor in reasoning quality.

- **Model Scale:** The 16GB VRAM limit necessitated the selection of BLIP-2-OPT-2.7B as the intermediate fusion exemplar rather than larger 7B parameter alternatives, introducing a size confound discussed in Section 4.7.
- **Sample Size:** The sample size was capped at 1,000 examples to ensure consistent inference times within the runtime limits of the T4 environment.

4.5 Prompt Engineering and Normalisation

Standardising model outputs for automated scoring required an iterative prompt engineering process. The two architectures exhibited substantially different sensitivities to prompt formatting, which is itself a behavioural signal relevant to the framework’s predictions about interaction mechanism design rather than a mere implementation detail.

- **LLaVA Prompting:** LLaVA converged reliably on a binary yes/no format. Initial attempts using three class prompts, Entailment, Neutral, and Contradiction, resulted in label instability, where the model struggled to maintain the predicted label space under contradiction.
- **BLIP-2 Prompting:** BLIP-2 failed to produce parseable outputs using LLaVA’s binary prompt, frequently echoing the prompt text or defaulting to a single class prediction. Reliability was achieved only through a completion style prompt:

Question: {hypothesis} The answer is:

4.6 Metrics and Analysis Approach

The primary metric is per condition accuracy, calculated independently for the entailment and contradiction conditions. This enables the calculation of an accuracy gap for each model, which serves as a measure of reasoning symmetry under semantic conflict.

Statistical uncertainty is reported using 95% confidence intervals calculated with the normal approximation for binomial proportions. These confidence intervals indicate the precision of each point estimate and help assess whether the observed asymmetry in BLIP-2’s performance is meaningfully different from balanced behaviour.

Two sample proportion tests are used to assess whether the difference between models on each condition is statistically significant. While the experiment is not designed as a benchmark and does not claim population level effects, these tests provide a principled basis for the claim that the two models behave differently under the same conditions.

The results are analysed against H1 and H2 specifically, assessing whether observed behaviour is consistent with the framework’s predictions for fusion stage and interaction mechanism. The analysis does not seek to validate the framework as a universal theory, but to evaluate its utility as an analytical tool for interpreting multimodal behaviour under contradiction.

4.7 Methodological Limitations and Confounds

Several significant limitations and confounds must be acknowledged when interpreting the results of this exploratory evaluation.

4.7.1 Confounded Design Dimensions

The two selected models differ on both the Fusion Stage (S) and Interaction Mechanism (I) dimensions simultaneously. LLaVA utilizes late fusion with direct MLP projection, while BLIP-2 utilizes intermediate fusion with a cross-attention bottleneck. Consequently, any observed performance differences cannot be definitively attributed to one dimension over the other. The experiment identifies the combined effect of these design choices rather than isolating their individual contributions.

4.7.2 Scale and Backbone Mismatches

The models employ different language backbones, Vicuna-7B for LLaVA and OPT-2.7B for BLIP-2. The difference in parameter count and training data means the models possess different baseline linguistic reasoning capacities, which may interact with the fusion architecture. Specifically, LLaVA’s larger language model may provide greater capacity to compensate for contradictory signals independently of the fusion mechanism.

4.7.3 Prompt Asymmetry

The requirement for different prompt formats, direct instruction for LLaVA versus completion for BLIP-2, introduces a procedural confound. While necessary for data collection, this asymmetry means the models were not evaluated under identical linguistic conditions. The degree to which the observed behaviour is driven by the prompt format rather than the fusion architecture remains an open question.

4.7.4 Single Dataset and Task

The use of a single dataset, SNLI-VE, and a single task, forced choice entailment, limits the generalisability of the findings. Multimodal reasoning behaviour may vary across different image domains, task formats, and contradiction types. The findings should be interpreted as indicative of behaviour on the SNLI-VE distribution rather than as a universal property of the architectures.

4.7.5 Quantization Effects

The use of 4-bit quantization may interact with fusion architecture in unforeseen ways. It is plausible that the already compressed Q-Former bottleneck in BLIP-2 is more sensitive to the representational loss of quantization than LLaVA’s uncompressed projector. Without a full precision baseline, the precise contribution of quantization to the observed failure modes cannot be determined.

4.8 Transition to Findings

Despite these limitations, the experiment provides a controlled environment for observing how two distinct fusion paradigms respond to the same 1,000 examples of semantic conflict. The next chapter presents the results of this evaluation and interprets them through the lens of the proposed framework, focusing on the observed asymmetries in model behaviour.

Chapter 5

Findings

5.1 Quantitative Results

This chapter presents the results of the exploratory contradiction experiment with minimal interpretation. The central objective is to show how the two selected design exemplars behaved under the same controlled test conditions before those results are interpreted through the framework in Chapter 6. Accuracy values are reported as percentages, while differences between accuracies are reported in percentage points.

Table 5.1 summarises the primary accuracy results for each condition. Values in brackets are 95% confidence intervals. LLaVA shows high and relatively balanced performance across entailment and contradiction, while BLIP-2 shows a strong asymmetry with a large decline in entailment accuracy.

Table 5.1. Accuracy by condition and model. Values in brackets are 95% confidence intervals. Accuracy gaps are reported in percentage points.

Model	Entailment Accuracy	Contradiction Accuracy	Accuracy Gap
LLaVA-1.5-7B	93.8% [91.7%, 95.9%]	91.0% [88.5%, 93.5%]	+2.8 [-1.8, +7.4]
BLIP-2-OPT-2.7B	26.0% [22.2%, 29.8%]	83.2% [79.9%, 86.5%]	-57.2 [-64.3, -50.1]

5.2 Visual Summary of the Results

Figure 5.1 presents the accuracy estimates for the two models and the two evaluation conditions. The figure uses a common scale from 0% to 100% and includes 95% confidence-interval error bars, allowing both the magnitude and uncertainty of the estimates to be compared directly.

The figure shows that LLaVA’s entailment and contradiction results remain close together. By contrast, BLIP-2 displays a pronounced separation between the two conditions, with substantially lower entailment performance than contradiction performance.

This pattern is therefore not a generic drop in accuracy. It is a directional shift. BLIP-2 becomes much less able to confirm entailment while remaining comparatively

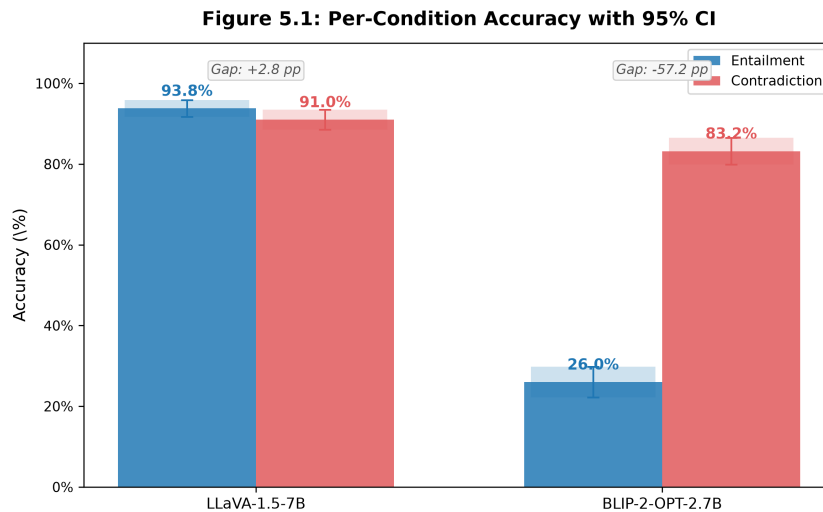


Figure 5.1. Accuracy by model and condition with 95% confidence intervals.

strong at rejecting contradiction.

5.3 Observed Failure Pattern

The accuracy gap provides a compact summary of the difference between the two conditions. LLaVA’s gap is close to zero, and its confidence interval crosses zero. This means that the observed difference is compatible with approximately balanced performance across entailment and contradiction.

BLIP-2’s gap is strongly negative, and its confidence interval lies entirely below zero. This indicates a strong and statistically stable asymmetry in favour of contradiction detection over entailment confirmation.

The complementary error-rate interpretation leads to the same conclusion. BLIP-2’s errors are concentrated in entailment rather than distributed evenly across both conditions. This reframing is useful because it draws attention to the failure mode rather than only to the success rate.

BLIP-2’s low entailment score is the most striking empirical outcome of the experiment. In practical terms, the model frequently treated entailment cases as contradiction cases, producing a pattern that resembles a default negative response under uncertainty. This is an observed phenomenon, not yet an explanation.

LLaVA did not show the same collapse. Its entailment and contradiction scores remained sufficiently close for the model to be described as broadly symmetric across the two conditions.

The between-model differences are also substantial. On entailment, LLaVA is far more accurate than BLIP-2, and the difference is statistically large rather than a product of sampling noise ($z = 30.29$, $p < 0.001$). On contradiction, both models perform relatively well, although LLaVA still maintains a smaller advantage ($z = 3.70$, $p < 0.001$).

The observed asymmetry should be read alongside the limitations in Chapter 4. The models differ in backbone size, prompt format, and quantisation sensitivity, so the result cannot be attributed to fusion stage alone. Even so, the outcome provides evidence that the overall fusion design choice is associated with materially different contradiction-handling behaviour.

5.4 Qualitative Examples

The quantitative pattern is supported by the qualitative behaviour observed during testing. LLaVA was generally able to return a stable binary answer after the prompt format was normalised, while BLIP-2 required more careful prompting before it produced parseable outputs at all. This prompt sensitivity suggests that the two architectures do not expose the same decision pathway to the input prompt.

The most useful qualitative cases are those where the two models diverged on the same image-hypothesis pair. In those cases, LLaVA more often aligned with the expected label, while BLIP-2 more often drifted towards contradiction even when the hypothesis was supported by the image. The point here is not to claim that every example behaved this way, but to show that the observed failure mode was not isolated to a single input.

A second useful qualitative observation is that BLIP-2’s behaviour changed substantially with prompt form. This means that the model’s output is sensitive not only to the multimodal content but also to how the task is framed linguistically. This supports the claim that prompt design and fusion design interact, which is one reason the experiment should be treated as exploratory rather than as a benchmark.

5.5 Behavioural Profile Comparison

The quantitative results can also be summarised through an accuracy-symmetry measure. Accuracy symmetry is defined as the complement of the absolute difference between entailment and contradiction accuracy:

$$S = 100\% - |A_{\text{entailment}} - A_{\text{contradiction}}|. \quad (5.1)$$

Here, the accuracies are expressed on the percentage scale. A score closer to 100% indicates more balanced performance across the two conditions. On this measure, LLaVA has a symmetry score of 97.2%, whereas BLIP-2 has a score of 42.8%.

This derived measure summarises the same contrast shown in Table 5.1 and Figure 5.1. It should not be treated as an additional independent metric. Prompt handling is discussed qualitatively because the experiment only supports an observational comparison, namely that LLaVA produced stable binary outputs after prompt normalisation, while BLIP-2 required more careful prompting before it produced parseable outputs.

The resulting profile remains descriptive rather than inferential. LLaVA displays a more balanced pattern across the two conditions, while BLIP-2 remains skewed

towards contradiction detection and lower entailment accuracy.

5.6 Limitations and Future Work

The experiment was designed as an exploratory evaluation with two models. Several limitations should be noted:

Exemplar selection. Only two models were tested: LLaVA, representing a late-fusion design, and BLIP-2, representing an intermediate-fusion design. No early-fusion exemplar, such as a model that merges modalities at the input level or uses early cross-attention, was included. The absence of an early-fusion data point means that the framework’s prediction about early fusion remains untested. An early-fusion exemplar may strengthen, refine, or challenge the current observations. Consequently, evaluating additional architectures remains an important direction for future work.

Confounds. The models differ in multiple dimensions beyond fusion stage, including language-model backbone size, with 7B parameters for LLaVA and 2.7B parameters for BLIP-2, quantisation level, and prompt format. These confounds are documented in Chapter 4 and should be considered when interpreting the results.

Dataset. The experiment used a 1,000-sample subset of SNLI-VE. Larger and more diverse datasets may reveal different patterns.

Prompt sensitivity. BLIP-2’s performance varied with prompt format, which suggests that the comparison is not fully controlled for prompt design. This is an inherent challenge when comparing architectures with different input-handling constraints.

5.7 Summary of Key Observations

The main findings can be summarised cautiously as follows:

- LLaVA shows near-symmetric performance across entailment and contradiction, with a small accuracy gap whose confidence interval crosses zero.
- BLIP-2 shows a pronounced asymmetry, with entailment accuracy of 26.0% and contradiction accuracy of 83.2%. Its accuracy-gap confidence interval lies entirely below zero.
- The asymmetry is directional rather than a general reduction in performance, because BLIP-2 remains comparatively strong on contradiction while performing substantially worse on entailment.
- The between-model differences are statistically significant for both entailment and contradiction, although the difference is much larger for entailment.
- Prompt-format sensitivity differs across the two architectures, with BLIP-2 requiring more careful prompt engineering.

- The accuracy-symmetry measure provides a compact summary of the contrast between LLaVA’s balanced profile and BLIP-2’s contradiction-biased profile.
- The results are consistent with, but do not prove, the view that fusion design choices influence contradiction handling.

These findings are intentionally descriptive. Chapter 6 now turns to interpretation, linking the observed symmetry, asymmetry, and prompt sensitivity to the framework’s fusion-stage and interaction-mechanism claims while keeping the current confounds explicit.

Chapter 6

Interpretation and Discussion

This chapter interprets the findings from Chapter 5 through the conceptual framework developed in Chapter 3. Chapter 3 proposed that multimodal reliability emerges from interacting architectural design choices rather than from any single component in isolation. The framework is presented as an original synthesis of existing perspectives on multimodal fusion, rather than as a new fusion mechanism or a validated predictive theory. Its contribution lies in organising several architectural dimensions within a single analytical structure for examining their possible relationship with reliability behaviour. The experiment provides a targeted empirical examination of the fusion-stage and interaction dimensions only; it does not validate the framework as a whole or establish causal mechanisms. The discussion below therefore distinguishes between observed findings, tentative interpretations, and directions for future research.

6.1 Framework Interpretation

The results are broadly consistent with the framework’s expectation that different positions in the (S, I) design space may produce different behavioural profiles. Chapter 5 showed that LLaVA (Liu et al., 2023) remained near-symmetric across entailment and contradiction, while BLIP-2 (Li et al., 2023) exhibited a marked asymmetry with entailment collapse.

6.1.1 Evaluation of H1, the Stage Hypothesis

The pattern is consistent with H1, which predicted that intermediate fusion would be more vulnerable to contradiction than late fusion. In the experiment, the intermediate-fusion model showed substantially larger degradation under contradiction than the late-fusion model. This does not by itself establish causation, but it does suggest that fusion stage is a plausible explanation for the observed asymmetry.

6.1.2 Evaluation of H2, the Interaction Hypothesis

The findings also speak to H2, but more cautiously. H2 anticipated that cross-attention-based interaction could reduce hallucination under contradiction by allowing selective weighting of modality-specific signals. BLIP-2’s behaviour, however, was not consistent with a simple protective effect of interaction mechanism alone, because it still showed a high entailment error rate. The more careful interpretation is that interaction mechanism may matter, but its effect appears to depend on the broader fusion design in which it is embedded.

6.2 Novelty and Contribution of the Framework

The framework does not claim that contradiction handling, hallucination, or robustness are newly identified multimodal phenomena. Its claimed contribution is instead to organise these related reliability concerns as interconnected consequences of architectural design choices. In this sense, the framework provides an original analytical synthesis rather than a new fusion mechanism or a fully validated predictive model.

The multiple views presented in Chapter 5, including accuracy comparisons, gap plots, error rates, and behavioural profiles, each isolate a different facet of the same underlying architectural pattern. Taken together, these views illustrate the framework’s value as a structured interpretive lens for examining why similar multimodal models may behave differently under conflict. They do not, however, independently validate the framework or establish that fusion stage is the sole cause of the observed differences.

6.3 Implications for Multimodal Evaluation

The findings have implications for how multimodal systems are evaluated. Aggregate metrics such as overall accuracy can conceal architecture-specific failure patterns. In this study, BLIP-2’s overall accuracy (0.546) could obscure a substantial difference between its entailment and contradiction performance if the two conditions were not reported separately.

The result illustrates the value of analysing model behaviour under contradiction rather than relying only on aggregate performance. Stress-testing models under conflicting multimodal conditions can help identify whether their responses remain balanced across conditions or are disproportionately influenced by one type of evidence.

6.4 Possible Bottleneck Effect and Language-Prior Interpretation

One possible interpretation is that BLIP-2’s Q-Former bottleneck may reduce the amount of visual information reaching the language model. In the experimental

setup, BLIP-2 compresses 196 visual patches into 32 query tokens, which may leave the downstream model with a more limited representation to work from. Under this interpretation, the entailment collapse is consistent with a situation in which compressed visual evidence is harder to use when the task requires positive support from the image.

This interpretation remains tentative because the experiment does not measure the model’s internal representations, attention allocation, or language-prior use. It therefore cannot determine whether the observed pattern resulted from information compression, a language prior, a negative default, or another decision process. Some studies have linked multimodal hallucination to reliance on language priors rather than visual input, particularly under conditions of information bottleneck (Huang et al., 2024). The present results show only that entailment was substantially more difficult than contradiction for BLIP-2. A bottleneck-based explanation is therefore plausible, but not established.

6.5 Confounds and Generalisability

The architectural interpretation should remain tentative because the two models differed in more than fusion stage. The study also varied model size, with LLaVA using a 7B Vicuna backbone and BLIP-2 using a 2.7B OPT backbone, and instruction tuning and prompt format also differed. These differences do not negate the framework interpretation, but they mean that the architectural reading is best treated as provisional.

Furthermore, the present findings should not be interpreted as showing that all intermediate-fusion architectures are less reliable than all late-fusion architectures. Rather, the experiment demonstrates that analysing architectures through fusion stage provides a useful explanatory lens for understanding behavioural differences under contradiction. Additional architectures and larger-scale testing are needed to determine how broadly these relationships generalise across the wider multimodal design space.

6.6 Industry Trends and Wider Context

As contextual background, recent multimodal architecture discussions have considered alternatives to restrictive bottleneck compression, including designs that preserve more visual information during integration. This observation is included as wider context rather than as an empirical finding of the present study. It illustrates that the trade-off between information compression and computational efficiency remains relevant to multimodal architecture design.

6.7 Implications for Untested Framework Dimensions

Because the experiment directly tested only the stage and interaction dimensions, no empirical conclusions can be drawn about representation, adaptivity, or compu-

tational cost. These dimensions are therefore better treated as directions for future investigation. In particular, future studies could examine whether manipulating information compression, confidence-weighted fusion, or computational constraints changes contradiction handling while holding other architectural factors constant.

6.8 Framework-Derived Design Directions

The findings also motivate several design proposals that extend the interpretation of the tested S/I pattern to the untested R , A , and C dimensions. These proposals were not implemented or evaluated in the present study and should therefore be treated as hypotheses for future testing rather than as validated solutions:

- **Contradiction-Aware Fusion Gates:** Dynamically adjust the depth of cross-modal interaction when semantic conflict is detected.
- **Adaptive Bottleneck Width:** Implement variable compression ratios that expand when input uncertainty is high.
- **Modality Confidence Scoring:** Assign per-instance reliability weights to modalities before integration to reduce the risk of overtrusting dominant signals (Han et al., 2022).

Detailed empirical evaluation of these proposals is beyond the scope of the present study and is considered as part of the future research agenda.

6.9 Synthesis and Conclusion

The central empirical pattern is clear, but the interpretation remains cautious. LLaVA exhibited more balanced behaviour across entailment and contradiction, while BLIP-2 exhibited a strong entailment collapse.

Rather than treating contradiction as an isolated failure mode, this thesis argues that it can be examined as a possible consequence of architectural design choices. The proposed framework therefore provides an analytical basis for interpreting observed reliability behaviours and formulating hypotheses about future multimodal architectures. By treating reliability as a property influenced by how information is integrated, this work provides a basis for further investigation into more robust and balanced multimodal systems.

Chapter 7

Conclusion

This chapter closes the thesis by consolidating its contributions, clarifying its scope, and identifying the directions it opens for future research. The preceding chapters built a conceptual framework (Chapter 3), tested two fusion architectures under controlled contradiction (Chapters 4 and 5), and interpreted the findings through the framework’s lens (Chapter 6). This final chapter synthesises those results into a concise statement of what was achieved, what was not, and what comes next.

7.1 Summary of Contributions

This thesis makes five principal contributions:

First, it proposes a conceptual framework that organises multimodal fusion design along five interacting dimensions: fusion stage (S), interaction mechanism (I), representation and alignment (R), adaptivity and reliability awareness (A), and computational cost (C). The framework is not a predictive model but an analytical lens that treats reliability phenomena such as contradiction handling, hallucination, and robustness as interconnected consequences of architectural design choices rather than isolated system failures.

Second, it provides empirical evidence that different positions in the fusion design space produce qualitatively different reliability behaviours. LLaVA, a late-fusion architecture, maintained near-symmetric accuracy across entailment (0.938) and contradiction (0.910) conditions. BLIP-2, an intermediate-fusion architecture with a Q-Former bottleneck, exhibited a catastrophic asymmetry: entailment accuracy collapsed to 0.260 while contradiction accuracy remained at 0.832, producing a 57.2-percentage-point gap. This pattern was statistically stable, with confidence intervals that did not overlap between conditions.

Third, it identifies the bottleneck effect and language-prior bias as plausible mechanisms linking fusion architecture to asymmetric reasoning failure. BLIP-2’s compression of 196 visual patches into 32 query tokens may reduce the visual information available to the downstream language model, which then defaults to its strongest internal prior (contradiction) when visual evidence is insufficient to

confirm entailment. This interpretation remains tentative because the experiment does not expose internal reasoning processes, but the observed pattern is consistent with this mechanism.

Fourth, it evaluates all three hypotheses against the empirical findings. H1 (Stage Hypothesis) is supported: intermediate fusion was more vulnerable to contradiction than late fusion. H2 (Interaction Hypothesis) is partially supported: cross-attention did not uniformly improve contradiction handling but appeared to interact with the broader fusion design. H3 (Adaptivity Hypothesis) was not directly tested, since no model with explicit confidence weighting was evaluated. The findings are consistent with the motivation behind H3, in that BLIP-2’s failure mode is the kind of behaviour reliability-aware fusion is intended to prevent, but this is illustrative motivation, not empirical support.

Fifth, it derives design implications for future multimodal architectures. The framework and findings suggest three concrete strategies: contradiction-aware fusion gates that dynamically adjust interaction depth when semantic conflict is detected, adaptive bottleneck widths that expand compression ratios when input uncertainty is high, and modality confidence scoring that assigns per-instance reliability weights before integration.

7.2 Remaining Limitations

Several limitations of this work remain beyond those already discussed in the context of the experimental design (Chapter 5) and the interpretation (Chapter 6).

Framework scope. The framework is a conceptual heuristic, not a validated theory. It does not account for training dynamics, data distribution, scaling laws, or fine-tuning strategies, all of which may influence how fusion design choices manifest in practice. Real multimodal systems resist clean categorisation along the five dimensions, and the framework’s value lies in its organising power rather than its predictive precision.

Dataset and task constraints. The experiment used a single dataset (SNLI-VE) and a single task (binary visual entailment) with a 1,000-sample subset. The dataset’s domain specificity (Flickr30k images with crowd-sourced hypotheses) limits generalisability to other visual domains, contradiction types, or open-ended generation tasks. The absence of a neutral condition means the binary framing may have amplified default-class effects.

Model selection. Only two models were tested, and both are openly available architectures. The framework’s predictions about early-fusion and hybrid architectures therefore remain unvalidated by direct experiment.

Evaluation depth. The experiment used a single primary metric (accuracy) without confidence scores, entropy measures, attention attribution, or open-ended generation evaluation. Richer evaluation instruments would provide a more complete picture of how the two architectures process contradiction internally.

7.3 Open Questions and Future Work

The findings open several directions for future research:

Full factorial design. A 2×2 factorial design crossing fusion stage and interaction mechanism would disentangle their independent and combined effects. This would require at least four models occupying distinct positions in the (S, I) space, matched for language model size and training strategy.

Empirical validation of H3. The adaptivity hypothesis was evaluated only theoretically. A direct test would compare the same base architecture with and without confidence-weighted fusion, measuring whether reliability-aware mechanisms reduce entailment collapse under contradiction.

Expanded task and domain coverage. The experiment used forced-choice binary classification. Extending to open-ended generation (image captioning, visual question answering, report generation) would reveal whether the observed asymmetry generalises to tasks where models must produce text rather than select between options. Cross-domain replication using medical, satellite, or abstract imagery would test whether the framework’s predictions hold beyond naturalistic photographs.

Richer evaluation metrics. Adding confidence scores, predictive entropy, and attention attribution would provide insight into how models process conflicting signals internally, not just what they output. This would strengthen the mechanistic interpretation of the bottleneck effect.

Architectural prototyping. The design directions proposed in Chapter 6 (contradiction-aware gates, adaptive bottleneck width, modality confidence scoring) are speculative. Implementing and evaluating one or more of these proposals would move the framework from analytical tool to practical design guidance.

Scaling and training analysis. Investigating how model scale, pre-training data, and instruction tuning interact with fusion design would address the confounds that limit the current study’s causal claims. A controlled ablation across model sizes within the same architecture family would be particularly informative.

This thesis set out to ask whether multimodal reliability could be understood as a consequence of architectural design choice rather than as an isolated, model-specific defect. The evidence gathered here, a stable, statistically distinguishable asymmetry in one architecture and its absence in another, is consistent with that view, though it cannot establish it conclusively from two confounded exemplars. What the work offers, more durably than any single empirical result, is a vocabulary for asking the question precisely: which dimension of fusion design is implicated, and what would have to be true for that implication to hold. Future work, outlined above, exists to test that vocabulary against more architectures, more tasks, and more direct measurement of the mechanisms proposed here.

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